CSA1701 - ARTIFICIAL INTELLIGENCE

7. Question: Write a Python program to implement BFS.

Aim: To traverse a graph using Breadth-First Search, visiting nodes level by level.

Algorithm:

1. Create a graph using an adjacency list.
2. Insert the starting node into a queue.
3. Remove the first node from the queue and print it.
4. Add its unvisited neighboring nodes to the queue.
5. Repeat until the queue is empty.

Program:

7 python.py - C:/Users/praji/AppData/Local/Programs/Python/Python314/7 python.py (3.14.8)

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```
from collections import deque

graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": [],
    "F": []
}

def bfs(start):
    visited = {start}
    queue = deque([start])

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for neighbor in graph[node]:
            if neighbor not in visited:
                visited.add(neighbor)
                queue.append(neighbor)

bfs("A")
```

IDLE Shell 3.14.8

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```
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/praji/AppData/Local/Programs/Python/Python314/7 python.py =
A B C D E F
>>>
```

Result: Thus, Breadth-First Search was implemented successfully.